

Must Read Before Use

1、 Calibrate Robotic Arm Offset

Strictly follow the steps below.

1.1、 Use APP to Calibrate Robotic Arm Neutral Position

Access the container. If the APP is already running, open the mobile app YahboomRobot, enter the IP address, connect to the robotic arm, click Arm Settings - Servo Calibration, and follow the prompts to calibrate the neutral position of the robotic arm.

If the APP is not running, enter the following command in the container to start the APP, then proceed with calibration:

```
python3 /home/yahboom/APP_DOFBOT_PRO/YahboomArm.pyc
```

2.2、 Use Calibration Program to Calibrate Offset (Only required if adjusting the initial position of the robotic arm for 3D spatial grasping; skip the tutorial below for other cases)

If you are missing the package, you can download the source code from the online repository provided. Place the source code into ~/dofbot_ws/. If you already have this package, you can ignore this step.

```
colcon build --packages-select dofbot_sorting
```

Enter the following command to start the calibration program in the first container terminal:

```
ros2 launch dofbot_info camera_arm_kin.launch.py
```

Then go to the same container terminal and enter:

```
ros2 run dofbot_sorting get_offset
```

Next, secure the robotic arm on the map as shown in the figure below:



The program will recognize the machine code block, then press the spacebar to complete calibration.

If you need to adjust the height, the package name is `dofbot_sorting`. If using the default position, the package is `dofbot_sorting_3d`. For example:

If color sorting in 3D spatial grasping requires starting:

```
ros2 run dofbot_sorting_3d color_sorting
```

If you need to modify the initial posture of the robotic arm:

```
ros2 run dofbot_sorting color_sorting
```

Notice: This offset value only applies to the current recognition angle. If subsequent program development involves angle modifications, recalibration will be required. This calibration is only applicable for 3D spatial grasping